

ELEC/COEN 390

Engineering team design project

MD NAZMUL HOQ

KHADIZA SARWAR MOURY

CONCORDIA UNIVERSITY

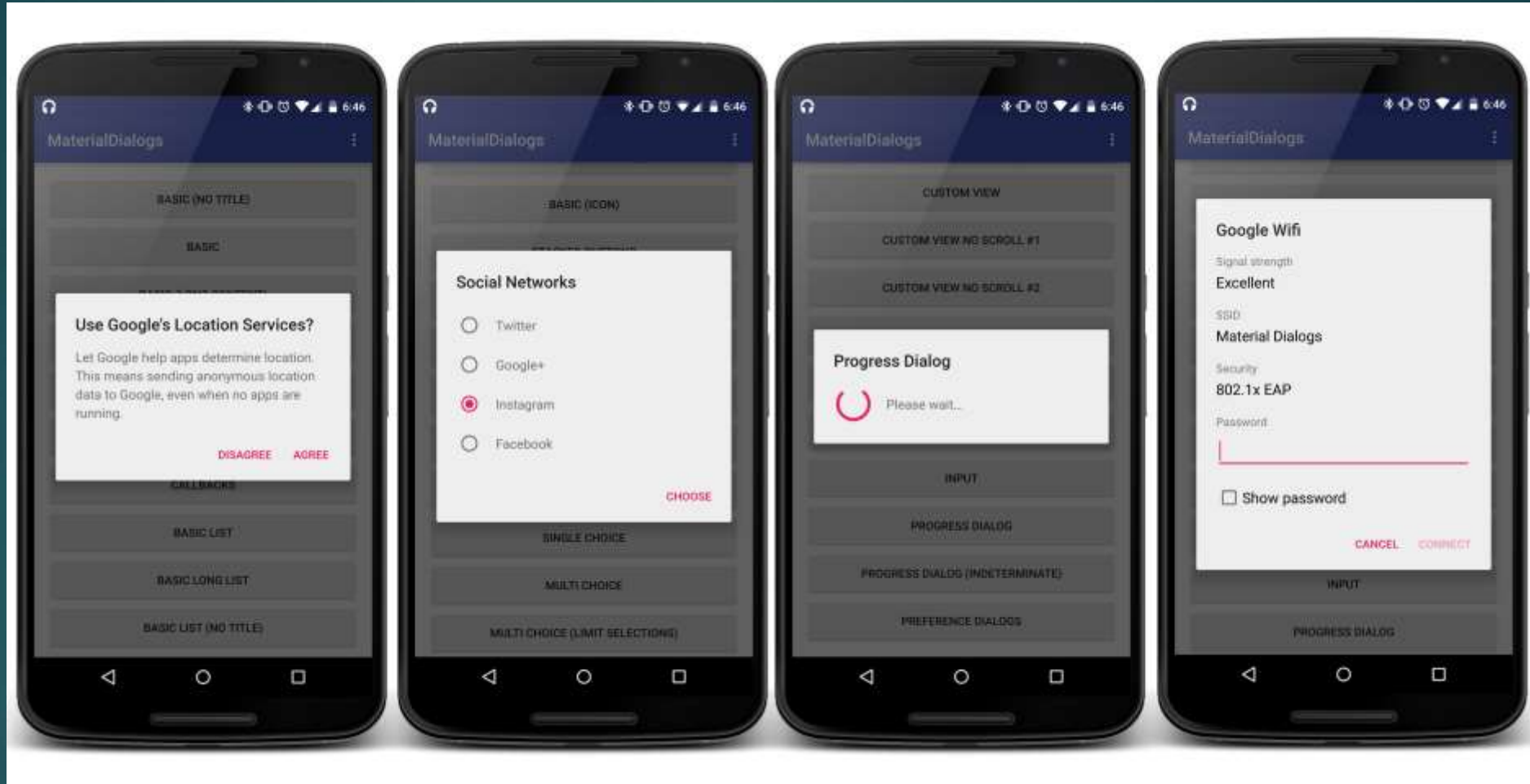
Understanding Fragments

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- ▶ A Fragment represents a **reusable** portion of your app's UI.
- ▶ A fragment defines and manages **its own layout**, has **its own lifecycle**, and can handle **its own input events**.
- ▶ Fragments **cannot live on their own**--they must be hosted by an activity or another fragment.
- ▶ Fragment reference:
 - ▶ <https://developer.android.com/guide/components/fragments>
- ▶ A Dialog Fragment is a specific type of fragment used to create floating popup dialogs
- ▶ DialogFragment reference:
 - ▶ <https://developer.android.com/reference/android/support/v4/app/DialogFragment>

DialogFragment examples

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► <https://guides.codepath.com/android/using-dialogfragment>

DEMO DialogFragment with SQL

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- ▶ After slides are done

GitHub with Android Studio

BASED ON SLIDES CREATED BY NICOLE CAPPADOCIA-ASSALY



Git Source Control

What you should use if you don't want to lose your code before
Demo Day

Typical Workflow

Pre-development:

1. Create repository.
2. Put into online repository.
3. Team clones the project.

During development:

1. Pull latest code.
2. Do the development, **and test that your changes didn't break anything.**
3. Pull the latest code, merge if necessary, and test that your changes didn't break anything.
4. Push to online repository.



Online Repositories

Repos, Private and Public

Repositories

- ▶ Git uses repositories to organize your code.
- ▶ You can have local repositories on your computer, or remote repositories online that you can share.
- ▶ Make sure to use a **PRIVATE** repo!
 - ▶ Github
 - ▶ Bitbucket
 - ▶ Gitlab
- ▶ Each team member needs to be a contributing member of the GitHub:
<https://github.com/username/projectname/settings/collaboration>

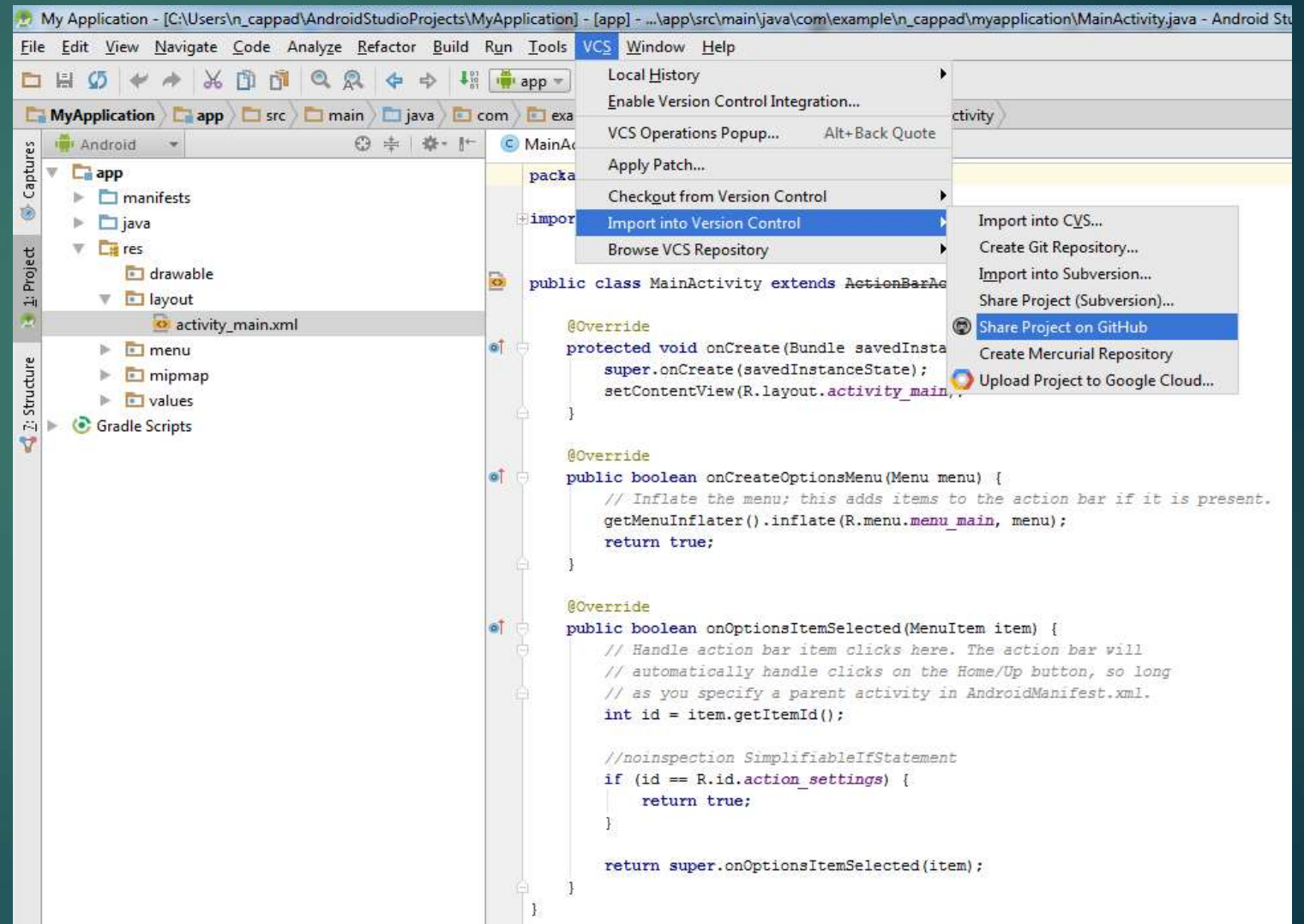
Create your Project Repo

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- ▶ Create your Android Studio Project

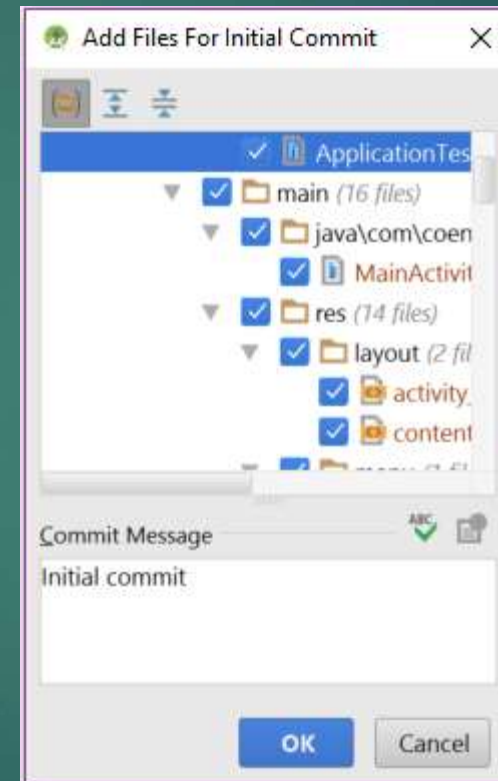
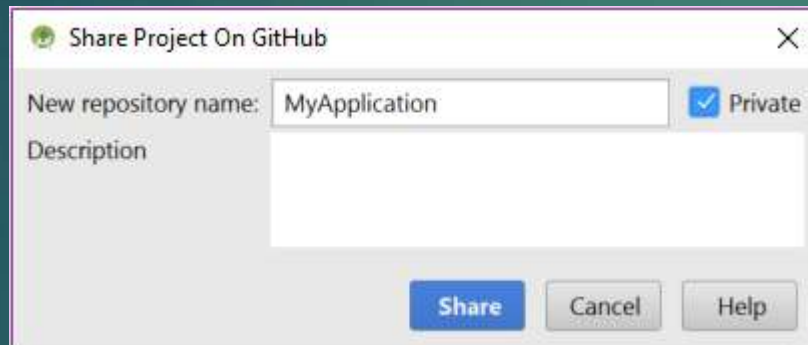
Then:

- ▶ **VCS > Import into Version Control > Share Project on GitHub**



Put it all in the cloud

- ▶ Enter password, set name and **PRIVATE**



- ▶ Add everything to your first commit.



Cloning

Getting someone else's existing code



Android Studio

Version 4.2.1

+ Create New Project

Open an Existing Project

Get from Version Control



Profile or Debug APK

Import Project (Gradle, Eclipse ADT, etc.)

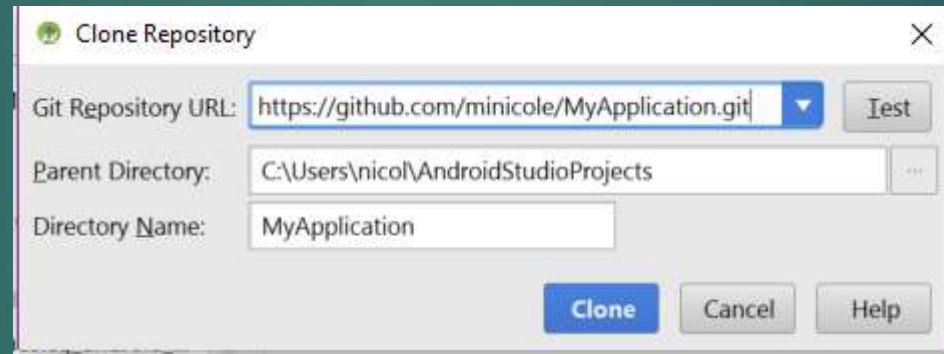
Import an Android Code Sample

Clone to Your Computer

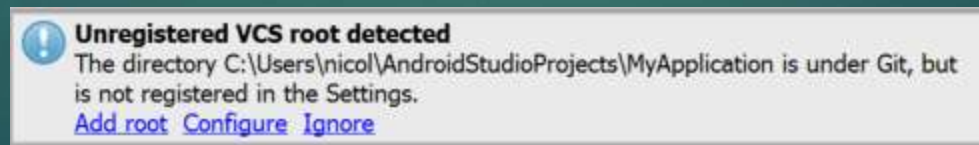
- ▶ After your project is in the online repo, you need to bring it to the team members' computers.
- ▶ Open Android Studio, and Get from Version Control

Clone to Your Computer

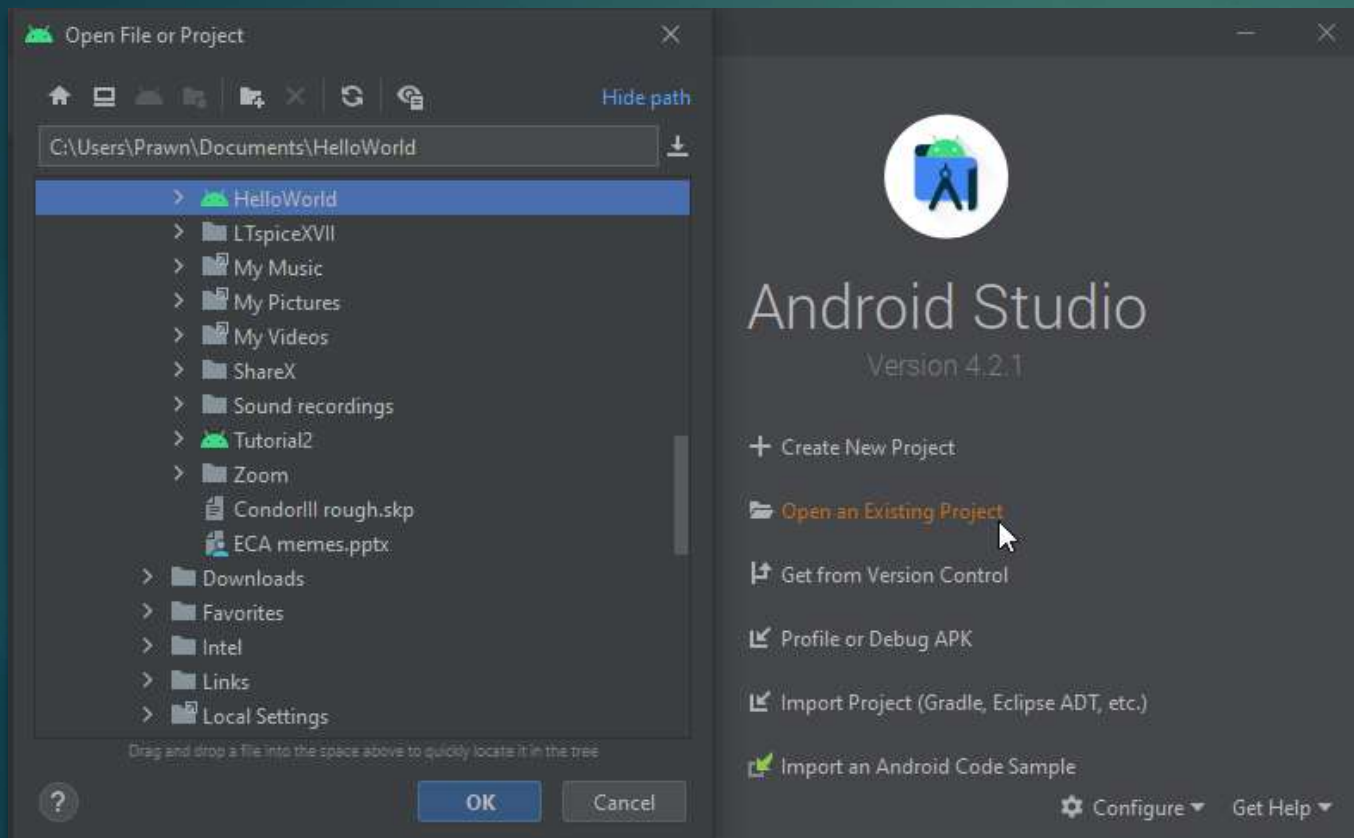
- ▶ Login to your GitHub account.
- ▶ Select your project.
- ▶ Clone!



- ▶ If there is a pop-up asking you to register VCS root, click Add Root.



Re-open Project



- ▶ If the project doesn't open properly, try closing the project, and then opening it as an existing Android Studio project.

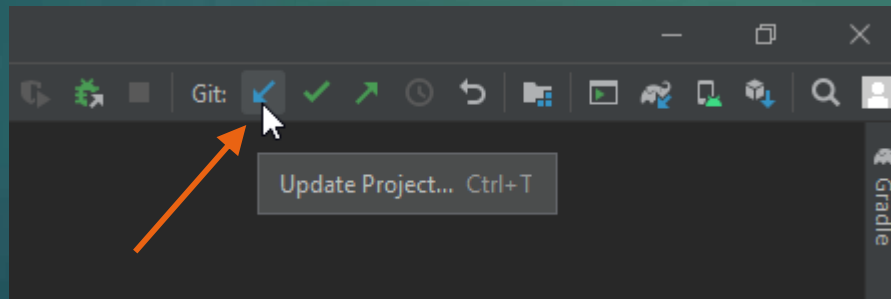


Update Project

The way to get new, up-to-date code

Update Project from the Online Repo

- ▶ Do before you start working (so that you don't do the same work as your colleagues)
- ▶ AND before you push (so that you don't overwrite your colleagues' work)
- ▶ Click on the Update Project button.
- ▶ This will bring the code changes from the online repo to your computer.
- ▶ Git Pull and Git Fetch can also be used





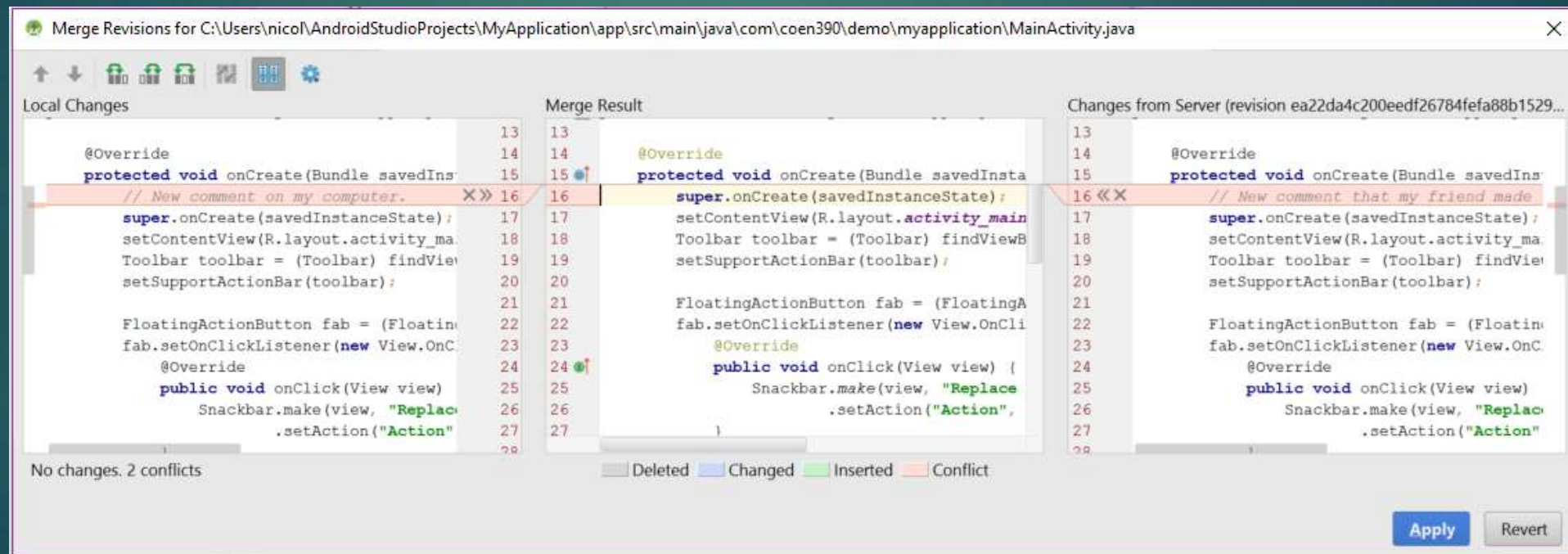
Merging

What you do when multiple users modified the same file

Merging

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- ▶ Git does auto-merging when possible.
- ▶ Otherwise, Android Studio prompts you to choose which lines to keep.



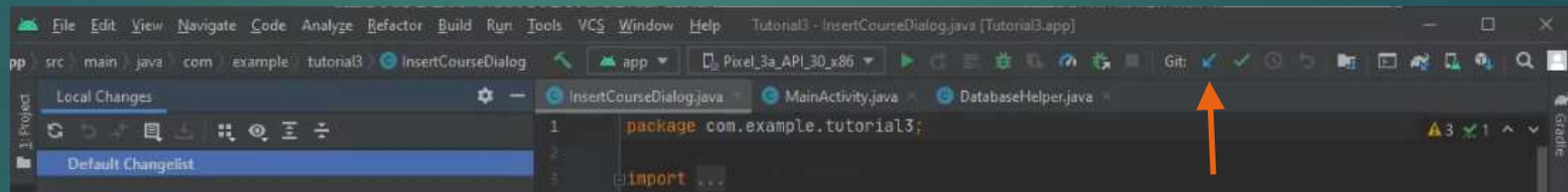


Committing & Pushing

Sending your code off to Github

Committing & Pushing

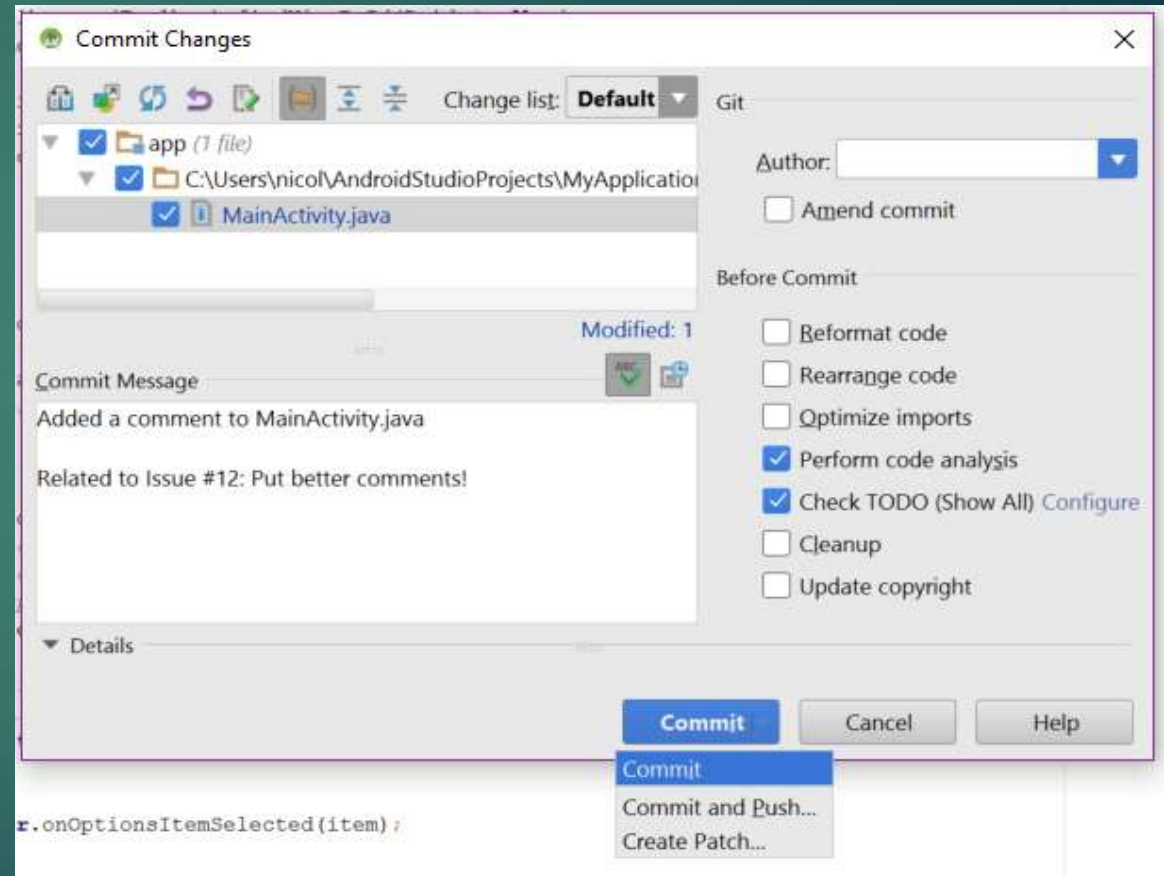
- ▶ When you are done working on an issue, commit your work.
- ▶ Commits are points of history in your project. (More on this later)
- ▶ Represent a snapshot of how your code is at a given time point.



- ▶ In Android Studio, just click on the Commit Changes button

Committing & Pushing

- ▶ A window will pop up where you choose the files to commit and can place a commit message
- ▶ All changed files selected by default
- ▶ Make your message meaningful and informative



Committing & Pushing

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	COMMENT	DATE
○	CREATED MAIN LOOP & TIMING CONTROL	14 HOURS AGO
○	ENABLED CONFIG FILE PARSING	9 HOURS AGO
○	MISC BUGFIXES	5 HOURS AGO
○	CODE ADDITIONS/EDITS	4 HOURS AGO
○	MORE CODE	4 HOURS AGO
○	HERE HAVE CODE	4 HOURS AGO
○	AAAAA	3 HOURS AGO
○	ADKFJSLKDFJSDKLFJ	3 HOURS AGO
○	MY HANDS ARE TYPING WORDS	2 HOURS AGO
○	HAAAAAANDS	2 HOURS AGO

AS A PROJECT DRAGS ON, MY GIT COMMIT MESSAGES GET LESS AND LESS INFORMATIVE.

Original comic from xkcd.com

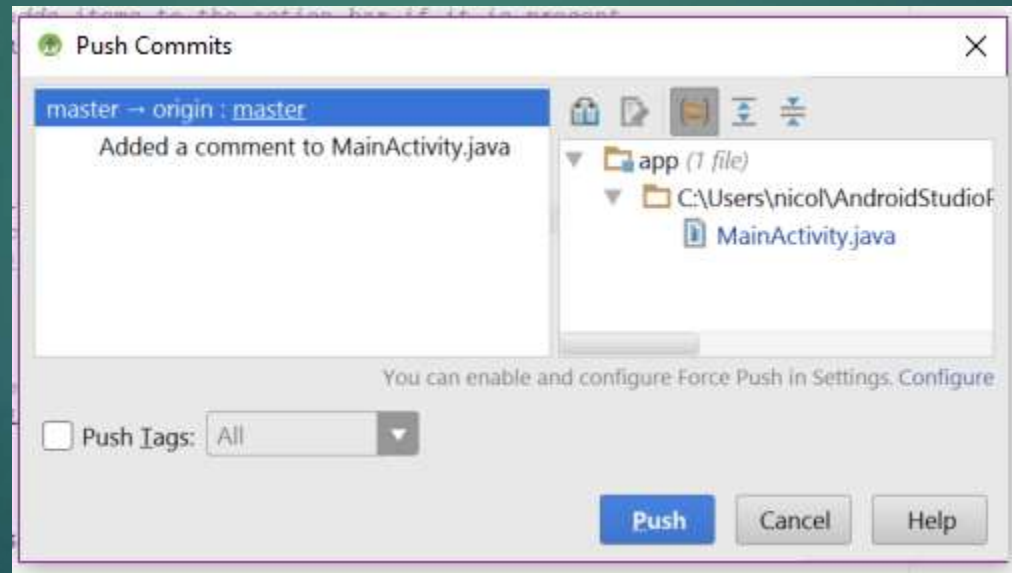
Don't just type anything.

Make good messages associated to your project items

Pushing

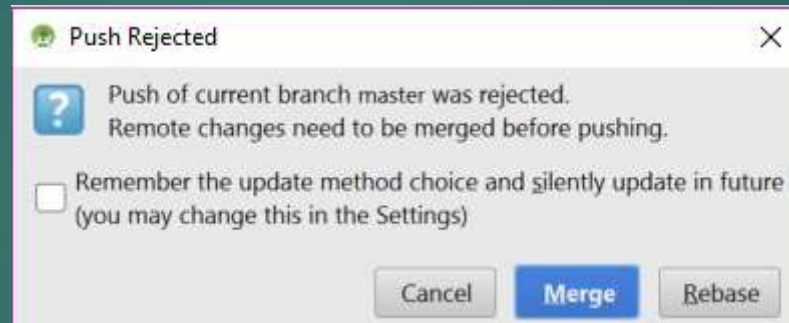
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- ▶ When you are ready to push to the remote repo, go to VCS > Git > Push... Or simply click on Commit and Push in the commit window (previous slide).
- ▶ This window will show you the commits that you are about to push.



Failure to Push

- ▶ If you forget to pull before you push, Android Studio might ask you to merge.



- ▶ Remember to your commit and the merge commit when you are done merging and after you have tested to make sure nothing broke in the merge.



Branching

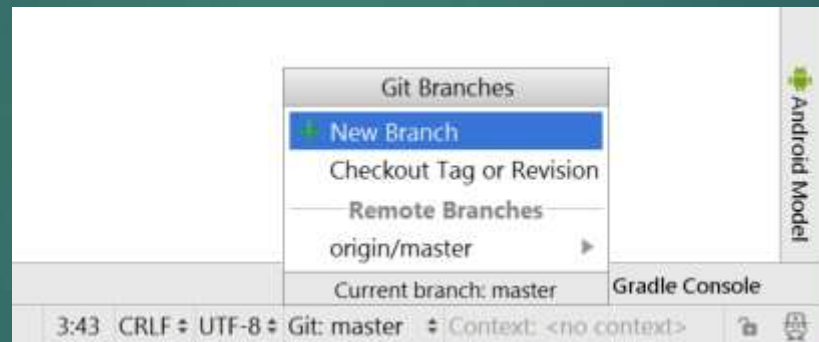
Work on something without causing a mess

Branching

- ▶ All repos come with a “Master” branch
- ▶ “Master” should always be stable and ready for release
- ▶ You’d “branch” from Master when:
 - ▶ You start a new task to work on something
 - ▶ The work on your feature might break other features
 - ▶ You need to work on two or more things at the same time, make a branch.
 - ▶ If you want to quickly see how a change would affect the master branch, make a branch.

How to Branch?

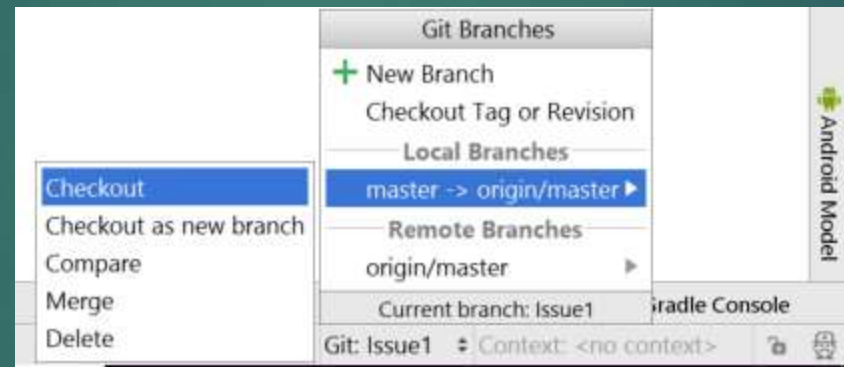
- ▶ At the bottom right corner of the screen, click on Git:currentBranch (master by default), then select New Branch, or an existing branch that you want to work on.



- ▶ Any changes that you had made that were not committed will follow you to the new branch.

How to go back?

- ▶ When you are ready to push to your master branch, click on master, and then click Merge.



- ▶ Then when you are sure that everything still works, you can switch to master using Checkout, and merge your branch to master.
- ▶ Git Checkout is also used to switch to any existing branch, or even individual commits

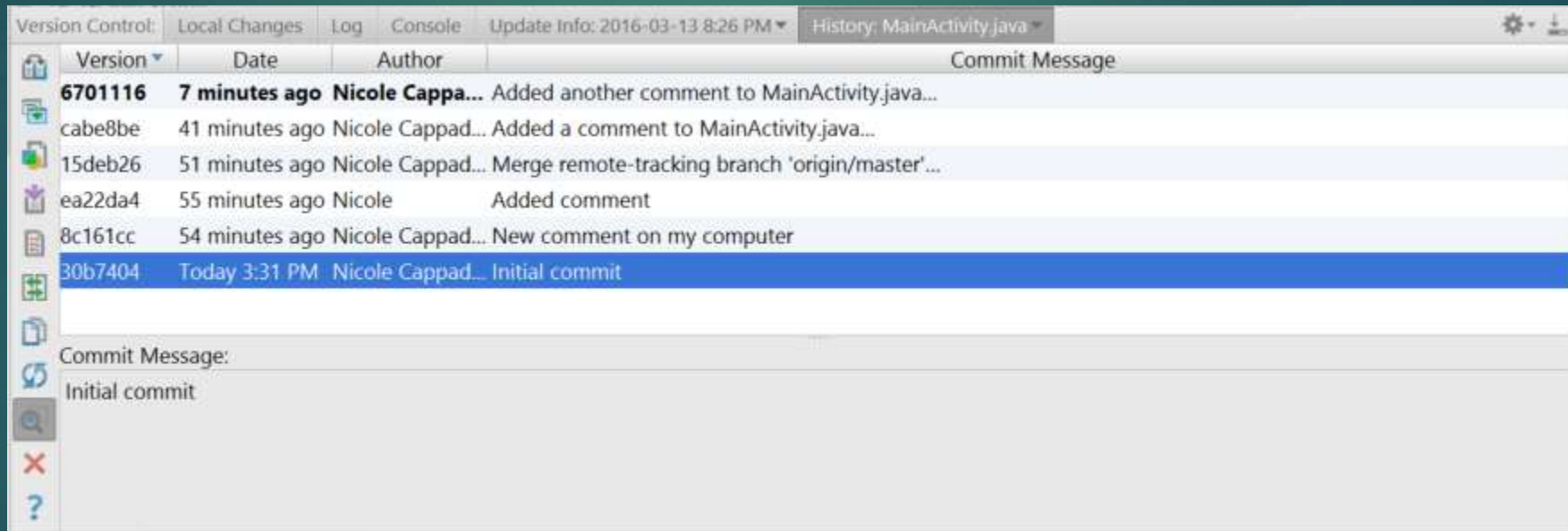


Reverting

What to do when you've made a mistake and things go wrong

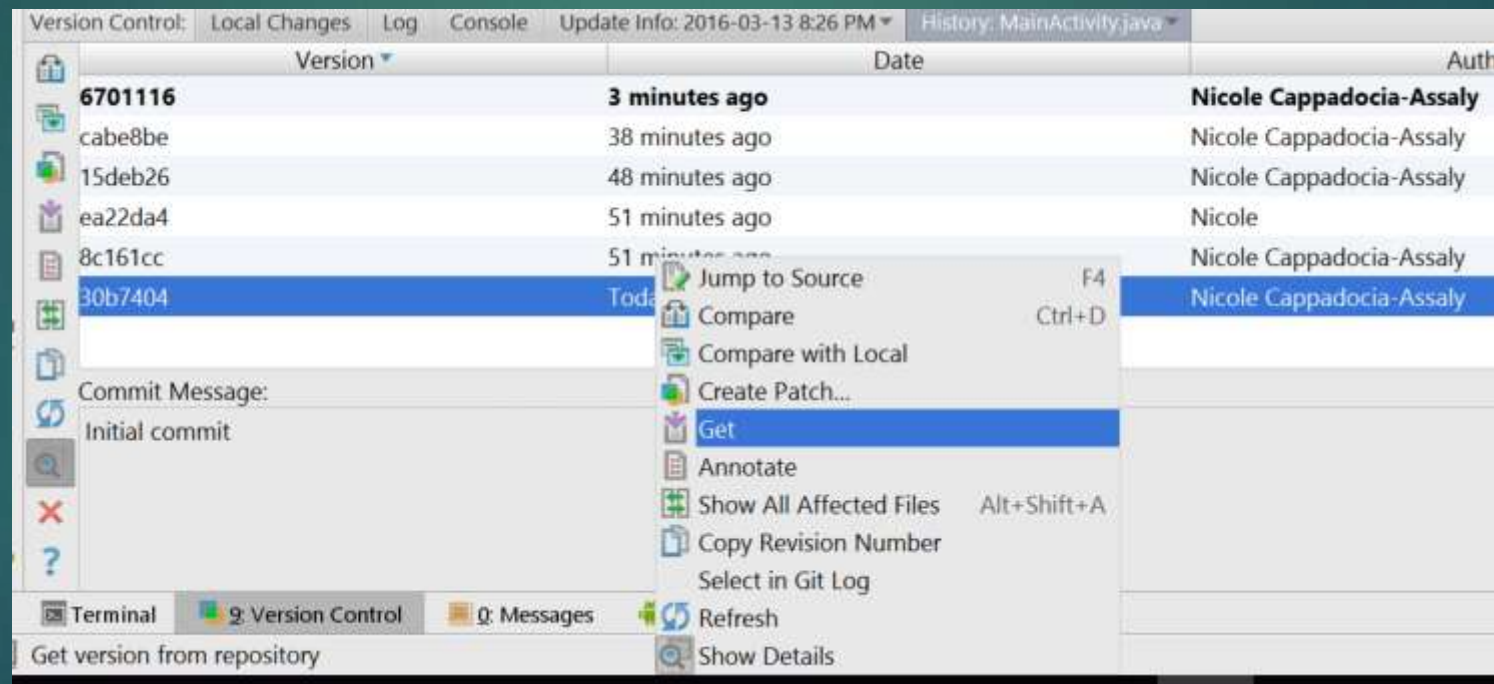
Seeing History

- ▶ Go to VCS > Git > Show History
- ▶ Here you will see each of the commits on the current branch.



Glance Back or Jump Back

- ▶ If you want to just see what the file looked like, click Jump to Source.
- ▶ If you want to return to that point in time, click Get.





View Online

What Github add on top of git

GitHub

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- ▶ Go to: <https://github.com/username/projectname/>
- ▶ You will see all the files, on all the branches.

The screenshot shows the GitHub interface for a repository named 'MyApplication' by user 'minicole'. The repository is marked as 'PRIVATE'. At the top, there are buttons for 'Unwatch' (1), 'Star' (0), and 'Fork' (0). Below this is a navigation bar with links for 'Code', 'Issues' (0), 'Pull requests' (0), 'Wiki', 'Pulse', 'Graphs', and 'Settings'. A message states 'No description or website provided. — Edit'. Below this, statistics show '6 commits', '1 branch', '0 releases', and '1 contributor'. A bar contains the current branch 'master', a 'New pull request' button, and buttons for 'New file', 'Upload files', 'Find file', and 'HTTPS' with the URL 'https://github.com/minico'. There are also icons for cloning and a 'Download ZIP' button. The commit history shows a recent commit by 'minicole' with the message 'Added another comment to MainActivity.java' 9 minutes ago. Below this, a list of files is shown: '.idea' (commented on computer, an hour ago), 'app' (commented on MainActivity.java, 9 minutes ago), 'gradle/wrapper' (initial commit, 5 hours ago), and '.gitignore' (initial commit, 5 hours ago).

minicole / MyApplication PRIVATE

Unwatch 1 Star 0 Fork 0

Code Issues 0 Pull requests 0 Wiki Pulse Graphs Settings

No description or website provided. — Edit

6 commits 1 branch 0 releases 1 contributor

Branch: master New pull request New file Upload files Find file HTTPS https://github.com/minico Download ZIP

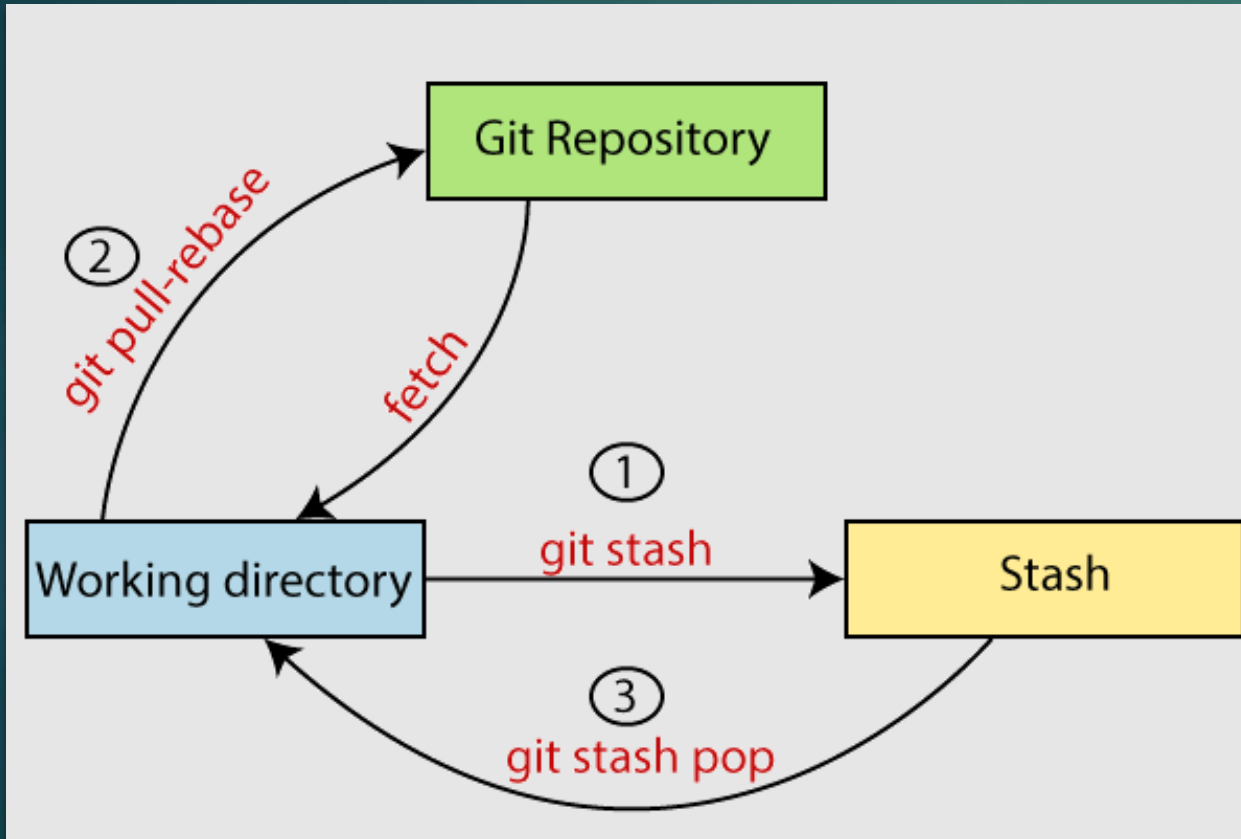
minicole Added another comment to MainActivity.java Latest commit 6701116 9 minutes ago

.idea	New comment on my computer	an hour ago
app	Added another comment to MainActivity.java	9 minutes ago
gradle/wrapper	Initial commit	5 hours ago
.gitignore	Initial commit	5 hours ago

GitHub Extras

- ▶ Here are some useful GitHub features that you should look into:
 - ▶ Pull Requests
 - ▶ Code reviewing by people in the group before merging to master branch
 - ▶ This is common software engineering practice
 - ▶ Graphs
 - ▶ See each user's contributions
 - ▶ Review what times were the most productive
 - ▶ Extrapolate Sprint speed
 - ▶ Issues
 - ▶ Project Management Tracking
 - ▶ [GitHub student package](#)
 - ▶ Free software and services for students

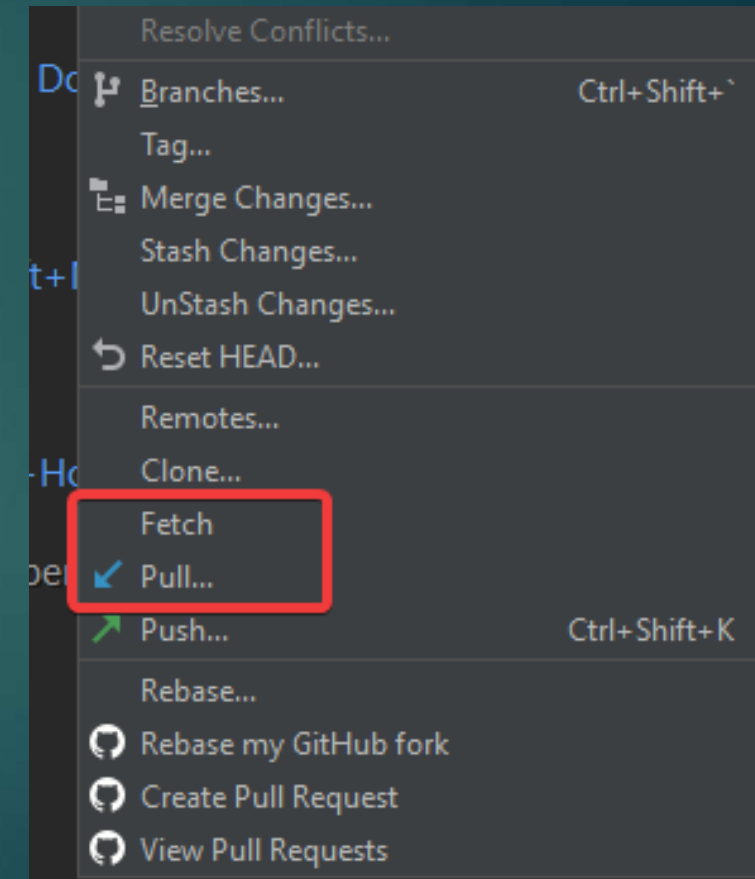
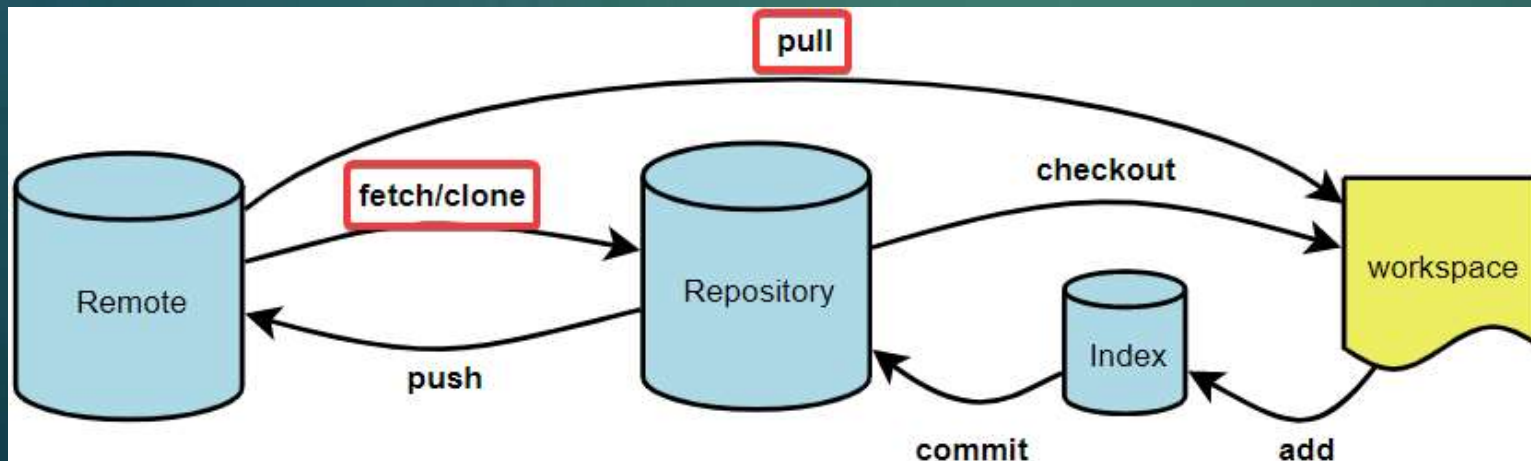
Additional git features



- ▶ Git stash & Git unstash
 - ▶ "Stash" changes away
 - ▶ Restores directory to last commit state
 - ▶ "unstash" retrieves the stashed changes

Additional git features

- ▶ Git Pull vs Git Fetch vs Update Project
 - ▶ Fetch only updates the local repo
 - ▶ Fetch does not merge to local workspace branch
 - ▶ Fetch is the “safest” option
 - ▶ Pull updates both repo & workspace branch
 - ▶ Pull is essentially “fetch + merge to local branch”
 - ▶ Update Project Pulls for all branches and roots



Questions?

DEMO DialogFragment with SQL